PSEUDO-COHERENCE OF A QUOTIENT OF A FLAT FINITELY PRESENTED ALGEBRA EQUALS PSEUDO-COHERENCE OF THE IDEAL

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 17517-17522 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 17524-17538, the complete author proof (13 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Algebraic hinge used by the chapter's relative pseudo-coherence chart checks (next wave's L16131 chart card builds on the same machinery).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma flat-finite-presentation-pseudo-coherent; More on Algebra, lemmas finite-push-pseudo-coherent and two-out-of-three-pseudo-coherent; the definition of pseudo-coherent ring map via polynomial presentations.

Source reference. Original lemma 0695: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17428–17430. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 064Z: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17538–17538. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 064V: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17538–17538. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $A \to B$ be a flat ring map of finite presentation. Let $I \subset B$ be an ideal. Then $A \to B/I$ is pseudo-coherent if and only if I is pseudo-coherent as a B-module.*

Proof. Choose a presentation $B = A[x_1, \ldots, x_n]/J$. Note that B is pseudo-coherent as an $A[x_1, \ldots, x_n]$-module because $A \to B$ is a pseudo-coherent ring map by Lemma 0695. Note that $A \to B/I$ is pseudo-coherent if and only if B/I is pseudo-coherent as an $A[x_1, \ldots, x_n]$-module. By More on Algebra, Lemma 064Z we see this

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is equivalent to the condition that B/I is pseudo-coherent as an B-module. This proves the lemma as the short exact sequence $0 \to I \to B \to B/I \to 0$ shows that I is pseudo-coherent if and only if B/I is (see More on Algebra, Lemma 064V). $\square$